01 - Represent a value with bits

Explained answers and feedback

Convert between unsigned binary, decimal and hexadecimal with a declared width.

Textbook: representation

A

58 = 32+16+8+2 = 00111010 = 3A. 10100110 = 128+32+4+2 = 166 = A6.

B

255 fits; 256 exceeds the maximum; -1 is outside the unsigned domain. A signed representation would require a different interpretation.

Feedback / exit

Leading zeros preserve an unsigned value; width establishes the available patterns/range. If a nibble is reversed, label bit weights before grouping into hexadecimal.

02 - Separate characters from encoded bytes

Explained answers and feedback

Distinguish a code point, encoding and the number of stored bytes.

Textbook: representation

A

41 42 C3 A9: 3 code points, 4 bytes, 32 bits. UTF-8 encodes the precomposed é using two bytes.

B

Ten precomposed é code points take 20 UTF-8 bytes. A displayed symbol can contain multiple code points, such as a base character plus a combining mark; do not equate every displayed symbol with one code point.

Feedback / exit

The bytes form a single UTF-8 sequence for U+00E9. Decode using the declared encoding. If totals count hexadecimal digits as bytes, mark pairs of digits as one byte.

03 - Combine logic and evaluate a small system

Explained answers and feedback

Construct a truth table and evaluate a declared access decision.

Textbook: selection

A

For staff/card/override in FFF, FFT, FTF, FTT, TFF, TFT, TTF, TTT order, allowed is F,F,F,F,F,T,T,T. TFF is false; TFT is true.

B

staff=False, card_valid=False, override=True: original false, changed expression true. The changed rule lets override bypass staff status as well as the card requirement.

Feedback / exit

Staff must be true and at least one of card_valid/override true. If combinations are missing, enumerate inputs as a three-bit count 000..111 before calculating output.

04 - Explain processing and a performance bottleneck

Explained answers and feedback

Use a simplified processing model and distinguish capacity from transfer rate.

Textbook: representation

A

Accept a labelled flow with sensor value as input, conversion/formatting as processing and displayed temperature as output. Saved program/log reside in persistent storage; program/data are loaded into working memory; immediate CPU values use registers. Working memory is generally volatile in this model.

B

Net growth 9-6 = 3 MB/s; fill time 24/3 = 8 seconds. A faster CPU does not necessarily increase the stated write throughput; storage remains the bottleneck unless its service rate changes.

Feedback / exit

MB measures an amount; MB/s measures an amount per second. Real rates vary, buffers may start partly full, and overhead may change throughput. Ask which assumption changed before recalculating.